

Bellona UAV — Detailed-Design Sizing Workflow (TU Delft DSE Group 22, Spring 2026)

TL;DR

- The detailed sizing workflow for the Bellona canard tail-sitter is built bottom-up from a 16-phase dependency graph (not the conventional "constraint-diagram-first" sequence); propeller and hover-power sizing precede the mission-trajectory optimisation that *derives* V_{cruise} and γ , which in turn feed an XFOIL airfoil loop and a wing-sizing step where V_{stall} is finally obtained from $CL_{\text{max},3D}$ and W/S — never from V_{cruise} .
 - The stability-and-control chain is reformulated around the *canard* neutral-point equation (Phillips, NASA TM-86694, 1985, Eq. 16: $x_{\text{np}}/\ell = [1 + a_w(1-\epsilon_{\alpha_c})S_w / a_c(1+\epsilon_{\alpha_w})S_c]^{-1}$), so the canard contribution shifts x_{np} *forward* (destabilising) — the opposite of the aft-tail formulation — and the elevon is sized by taking the maximum of the FW-roll case (Phase 11) and the hover-yaw-in-propwash case (Phase 12), with slipstream dynamic pressure $q_{\text{slip}}(\text{hover}) = T/A_{\text{disc}}$ from Stone 2008.
 - The constraint diagram (Raymer Ch. 5) is demoted to a verification check after Phases 3, 5 and 8; MTOW convergence is closed in a damped outer loop (Gundlach 2014, Ch. 19) with $|\Delta\text{MTOW}|/\text{MTOW} < 1\%$.
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Key Findings

1. **Sequence comes from the dependency graph.** Propeller diameter (Phase 1) depends only on MTOW seed and disc-loading regression; hover power (Phase 2) needs Phase 1 plus an ISA model at 6 km; the mission-trajectory optimisation (Phase 3) needs Phase 2 and an initial drag polar and *derives* V_{cruise} , γ , ROC and transition altitude as decision variables. Only then does the aerodynamics chain (Phases 7-9) start, with V_{stall} emerging from Phase 8 (not assumed in Phase 3).
2. **The canard neutral-point sign must flip.** Phillips 1985 Eq. 16 places x_{np} *between* the canard AC and the wing AC; increasing S_c or a_c moves x_{np} forward. Caughey (Cornell MAE 5070, 2011, Eq. 3.29) states this explicitly: ΔNP "is positive for conventional (aft tail) configurations, and negative for canard (forward tail) configurations." Using the aft-tail formula on Bellona would over-predict static margin by tens of percent of MAC and produce a fictitiously large allowable CG range. (cornell)
3. **Elevon sizing is binding in the hover-yaw case, not the FW-roll case.** With $V_{\infty} \rightarrow 0$ in hover, the freestream q vanishes and the only available dynamic pressure on the elevons is the propwash from the forward X-rotors: $q_{\text{slip}}(\text{hover}) = T/A_{\text{disc}}$ (from momentum theory

as applied by Stone 2008). For Bellona's baseline disc loading ($\sim 70\text{--}90 \text{ N/m}^2$ at $\text{MTOW}/n_{\text{rotors}} = 12.5 \text{ kg per rotor on } 0.93 \text{ m discs}$), this gives $V_{\text{slip}} \approx 12\text{--}14 \text{ m/s}$ in hover — comparable to V_{stall} , so the elevons must be large enough to produce the required yaw moment at this *fixed* propwash q , which typically dominates over the FW-roll case for X-config tail-sitters.

4. **Static-margin band is opened up by the autopilot.** Gundlach 2014 Ch. 10 and Beard & McLain 2012 Ch. 4–6 jointly justify $\text{SM} = 5\text{--}15\%$ MAC for autopilot-equipped small UAVs, with the lower bound permissible because the autopilot stabilises the short-period mode. This widens the scissor-plot feasible band and allows a smaller S_c than a manned-aircraft 10–15 % SM rule would require.
 5. **Battery first-guess specific energy is 200 Wh/kg at pack level — the *upper bound of current LiPo practice***, not a mid-range value (timharper.net 2025 "Battery Energy Density"; EcoFlow Oct 2025 verbatim: "their energy storage typically ranges from 150 to 250 Wh/kg for LiPo packs used in small multirotors"). With usable fraction $f_{\text{usable}} = 0.85$ (Gundlach Ch. 8), a 25-minute mission with $\sim 14 \text{ kWe}$ hover peaks demands a battery mass of $\sim 15\text{--}18 \text{ kg}$ — by itself $\sim 30\%$ of the 50 kg MTOW seed, so the MTOW outer loop is essential. (EcoFlow) (ecoflow)
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DELIVERABLE 1 — Full Workflow Document (16 phases)

Propulsion & Performance

Phase 1 — Physical propeller sizing. *Inputs:* MTOW seed (50 kg), $T/W = 1.30$ design / 1.15 floor (Stone & Clarke 2001), $n_{\text{rotors}} = 4$, X-config geometry envelope, ISA(6 km) $a = 316.4 \text{ m/s}$.

Governing equations:

- Disc-loading regression: $T/A = f(D)$ from Roskam Part VI Ch. 6 (Tab. 6.3) and Gundlach 2014 Ch. 7 (Eq. 7.18 form): $D_{\text{prop}} = \sqrt{(4 \cdot T_{\text{per rotor}} / (\pi \cdot DL_{\text{target}}))}$. Target $DL = 100\text{--}300 \text{ N/m}^2$ for VTOL UAVs (Gundlach Ch. 7).
- Tip-Mach constraint: $V_{\text{tip}} = \pi \cdot n \cdot D \leq 0.85 \cdot a_{\text{local}} \rightarrow n_{\text{max}} = 0.85 \cdot a / (\pi \cdot D)$. The 0.85 limit is from Hibbs (Kitplanes "Prop Blade Mach") and corroborated by Firgelli; thicker wooden blades use 0.8 — composite fixed-pitch blades for UAVs use 0.85 (Gudmundsson 2022 Ch. 14). (liminfo) (Kitplanes)
- Ground-clearance: tail-sitter standing on landing skids $\rightarrow$ minimum $D + \text{tip-to-ground gap} > 0.10 \text{ m} + \text{half-prop-clear-of-fuselage}$.
- X-config packaging: rotor-rotor separation $\geq 1.05 \cdot D$ (5 % gap to avoid disc-disc interaction, Gundlach Ch. 7); rotor-to-wing leading-edge $\geq 0.25 \cdot D$ for axial mixing (Stone 2008).
Outputs: D_{prop} (verify 0.93 m), n_{max} , rotor positions in body frame. *Handoffs:* Phase 2 (A_{disc}), Phase 4 (J target), Phase 12 (rotor moment arms). *Open assumptions:* DL_{target} value; fixed-pitch vs variable-pitch (assume fixed-pitch composite). *Why this phase first:* It

depends only on the MTOW seed and on top-level mission constants (T/W floor, n_{rotors}). No aerodynamic or stability output is needed.

Phase 2 — Hover and climb power at altitude. *Inputs:* D_{prop} (Ph 1), MTOW seed, ISA(6 km): $\rho = 0.660 \text{ kg/m}^3$, $a = 316.4 \text{ m/s}$; $\text{FoM} = 0.70$, $\eta_{\text{motor}} = 0.90$, $\eta_{\text{ESC}} = 0.95$. *Governing equations:*

- Ideal momentum theory induced velocity in axial *climb*: $v_i = -V_c/2 + \sqrt{((V_c/2)^2 + T/(2\rho A))}$ (Leishman 2006 Eq. 2.91, "Momentum Theory in Climb"). For static hover ($V_c = 0$): $v_h = \sqrt{T/(2\rho A)}$ (Leishman Eq. 2.42).
 - Ideal induced power in climb: $P_i = T(V_c + v_i)$ (Leishman Eq. 2.96).
 - Real shaft power: $P_{\text{shaft}} = T(V_c + v_i)/\text{FoM}$; electrical: $P_{\text{elec}} = P_{\text{shaft}} / (\eta_{\text{motor}} \cdot \eta_{\text{ESC}})$.
 - Show $P_{\text{climb}} / P_{\text{hover}}$ for $V_c = 0 \dots 5 \text{ m/s}$ — the climbing-flight correction reduces v_i but increases the $V_c \cdot T$ component, so net P typically rises $\sim 10\text{--}20\%$ above static hover.
- Outputs:* P_{hover} , $P_{\text{climb}}(V_c)$ curve, induced velocity v_h for slipstream calcs. *Handoffs:* Phase 5 (energy timeline), Phase 12 (v_h for elevon-in-propwash q). *Open assumptions:* $\text{FoM} = 0.70$ (small electric prop, Gundlach Ch. 7 Tab. 7.4 gives 0.65–0.75); ρ at intercept altitude only — for VTOL climb, density varies with altitude and the energy integral in Phase 5 must integrate $\rho(h)$. *Why now:* Hover power baseline is needed by the trajectory optimiser (Ph 3) and the energy timeline (Ph 5).

Phase 3 — Mission-trajectory optimisation: derive V_{cruise} , γ , ROC, $h_{\text{transition}}$. *Inputs:* $P_{\text{hover}}(h)$ and $P_{\text{climb}}(h, V_c)$ from Ph 2; initial drag polar $C_D = C_{D0} + C_L^2/(\pi A Re)$ with seed $C_{D0} = 0.04$, $e = 0.78$; MTOW seed. *Governing equations:*

- Power required in FW climb: $P_{\text{req}}(V) = \frac{1}{2}\rho V^3 SC_{D0} + 2W^2/(\rho VS\pi A Re)$ (Raymer 2018 Eq. 17.20).
- Speed for minimum power required (Raymer 2018 Ch. 17, derived from setting $dP_{\text{req}}/dV = 0$): $V_{P_{\text{min}}} = \sqrt{((2W/(\rho S)) \cdot \sqrt{K/(3C_{D0}))})}$ where $K = 1/(\pi A Re)$. The text observes that for an ideal constant- η_p propeller, max-rate-of-climb speed equals minimum-power speed; for a real propeller $V_{RC,\text{max}}$ is slightly higher because η_{prop} rises with V .
- Rate of climb: $\text{ROC} = (P_{\text{avail}} - P_{\text{req}})/W = V \cdot \sin \gamma$ (Raymer Eq. 17.22; Marchman 2022 Ch. 5).
- Total mission energy: $E_{\text{total}} = \int_0^{h_{\text{tr}}} P_{\text{hover_climb}}(h) dh / \text{ROC}_{\text{vert}} + \int_{h_{\text{tr}}}^{6000} P_{\text{req,FW}}(V, \gamma) / \text{ROC}_{\text{FW}} dh + P_{\text{hover}}(6000) \cdot 300 + E_{\text{tow}} + E_{\text{return}} + E_{\text{descent}} + E_{\text{reserve}}$.
- Optimise $h_{\text{transition}}$ by setting $\partial E_{\text{total}} / \partial h_{\text{tr}} = 0$ subject to (a) 10-min outbound time, (b) 6 000 m ground range, (c) $T/W \geq 1.15$ at h_{tr} (Stone & Clarke 2001). *Outputs:* V_{cruise} (= V at minimum-power point in level cruise), γ_{climb} , ROC, $h_{\text{transition}}$, Re_{estimate} for Phase 7. *Handoffs:* Phase 4 ($V_{\text{cruise}} \rightarrow J$), Phase 7 (Re_{estimate}), Phase 8 (V_{cruise} — but NOT V_{stall}). *Open assumptions:* Initial C_{D0} and e from baseline; will be refined when Phase 8

closes. Linear ISA interpolation for $\rho(h)$ over 0–6 km is acceptable here (ISA 1976 model).

Why now: Sets the FW operating point; the airfoil cannot be selected on Re without it.

Phase 4 — Dual-mode propeller coupling ($V_{\text{cruise}} \leftrightarrow J \leftrightarrow D_{\text{prop}}$ loop). *Inputs:* V_{cruise} (Ph 3), D_{prop} (Ph 1), n_{max} (Ph 1). *Governing equation:* $J = V_{\text{cruise}} / (n \cdot D)$. Target $\eta_{\text{prop}} > 0.7$ at $J \in [0.4, 0.8]$ for fixed-pitch electric props (Gudmundsson 2022 Ch. 14 §14.6; UIUC propeller database Selig 2002–2014). *Decision rule:* If $J(V_{\text{cruise}}, n_{\text{cruise}}, D) \notin [0.4, 0.8] \rightarrow$ revise D or n ; iterate. *Outputs:* Confirmed D_{prop} , RPM schedule for hover/climb/cruise/tow. *Handoffs:* Phase 1 if D revises (loop back), Phase 5 (cruise η_{prop}). *Why now:* J depends on V_{cruise} from Ph 3; must run before energy timeline.

Phase 5 — Motor, ESC, battery sizing — full mission energy timeline. *Inputs:* $P_{\text{segment}}(t)$ for each mission phase from Ph 2/3; η_{prop} schedule from Ph 4. *Governing equations:*

- Segment energies: $E_i = \int P_i dt$ (Nichols/Gundlach 2014 Ch. 8 Eq. 8.4).
- Battery sizing: $E_{\text{batt}} = \sum E_i / (\eta_{\text{batt}} \cdot f_{\text{usable}})$, with $\eta_{\text{batt}} = 0.95$ round-trip and $f_{\text{usable}} = 0.85$ (Gundlach Ch. 8 §8.6).
- Battery mass: $m_{\text{batt}} = E_{\text{batt}} / e_{\text{batt}}$ with $e_{\text{batt}} = 200 \text{ Wh/kg}$ (LiPo 4S–8S pack-level, upper bound of current practice; EcoFlow Oct 2025 verbatim: "their energy storage typically ranges from 150 to 250 Wh/kg for LiPo packs used in small multirotors"; timharper.net 2025 "Battery Energy Density 2025": "pack-level densities typically reach 150–200 Wh/kg after accounting for cooling, structure, and safety systems").
- Motor sizing: $P_{\text{motor,cont}} \geq P_{\text{climb}} / (\eta_{\text{ESC}})$; $P_{\text{motor,peak}} \geq P_{\text{hover_max}} (\sim 1.3 \times \text{steady hover for control margin}) / (\eta_{\text{ESC}})$. Gundlach Ch. 9 Tab. 9.3 gives motor-mass regression $m_{\text{motor}} \approx 0.27 \text{ g/W}$ continuous for outrunners at 100–1000 W.
- ESC continuous current: $I_{\text{ESC,cont}} = P_{\text{motor,cont}} / V_{\text{bus}}$ with $1.5 \times$ margin. *Outputs:* E_{batt} , m_{batt} , motor/ESC specs. *Handoffs:* Phase 15 (component masses), Phase 16 (MTOW iterate). *Open assumptions:* e_{batt} at the 50 kg-class is pack-level not cell-level; tow drag (894 N) and tow speed need to come from the mission spec (here taken at 13 m/s). *Why now:* Needs both Phase 3 (energy per FW segment) and Phase 2 (hover energy).

Phase 6 — Constraint diagram (VERIFICATION ONLY). *Inputs:* V_{cruise} , V_{stall} (from Ph 8), ROC, γ , T/W floor. *Governing equations:* Raymer 2018 Ch. 5 §5.3 — for propeller aircraft plot (W/P) vs (W/S): Scribd

- Stall: $W/S = \frac{1}{2} \rho V_{\text{stall}}^2 C_{L,\text{max}}$
- Cruise: $(W/P)_{\text{cruise}} = \eta_p / (V \cdot (C_{D0} \cdot \frac{1}{2} \rho V^2 / (W/S) + (W/S) / (\pi A Re \cdot \frac{1}{2} \rho V^2)))$
- Climb: $(W/P)_{\text{climb}} = \eta_p / (ROC + V \cdot D/W)$
- Transition floor: $T/W \geq 1.15$ (Stone & Clarke 2001).
- Hover: $T/W \geq 1.30$ (Bellona design). *Decision rule:* Confirm the derived (W/S, W/P) pair from Phases 3, 5, 8 falls inside all constraints. If not, flag for redesign; this is NOT a sizing driver

— sizing is already done. *Why this position:* It is a check at the end of Phases 3, 5, 8 — not first.

Aerodynamics

Phase 7 — Airfoil selection via XFOIL. *Inputs:* Initial Re from MTOW seed and first-guess W/S → V_{cruise} from Ph 3 → $Re_{\text{main}} = \rho V \bar{c} / \mu \approx 7.88 \cdot 10^5$ at 6 km; $Re_{\text{canard}} \approx 3.61 \cdot 10^5$. *Governing tools:* XFOIL (Drela 2001 *XFOIL User Guide*) with $N_{\text{crit}} = 9$ (free transition) and forced transition at $x/c = 0.50$ both surfaces (Selig 2003 VKI lecture notes for low-Re design). *Evaluation:*

- **Main wing:** maximise Cl_{max} and Cl/cd at design point; cm_0 small (low trim penalty); benign stall. SD7037 (Selig/Donovan, UIUC LSAT) is the reference; alternatives Eppler E216, NACA 4412, Wortmann FX 63-137. XFOIL predictions at $N_{\text{crit}}=9$ (UIUC database via airfoiltools.com) give SD7037 $Cl/Cd_{\text{max}} = 74.7$ at $\alpha = 4.75^\circ$ at $Re = 2 \cdot 10^5$, rising to $Cl/Cd_{\text{max}} = 99.3$ at $\alpha = 4.25^\circ$ at $Re = 5 \cdot 10^5$, with an interpolated value of ~ 82 at $Re = 3 \cdot 10^5$. The Cl_{max} of SD7037 "is not particularly sensitive to roughness" (Giguère & Selig 1997, *Wind Engineering* Vol. 21 No. 6, "Low Reynolds Number Airfoils for Small Horizontal Axis Wind Turbines" — note this insensitivity applies specifically to Cl_{max} , not to Cd or L/D).
([airfoiltools](http://airfoiltools.com)) ([UIUC Applied Aerodynami...](http://uiucappliedaerodynamics.org))
- **Canard:** gentle stall is the canard design rule (canard MUST stall before main wing for natural pitch-down recovery — Phillips 1985 §III, Roskam Part II Ch. 8). NACA 0012 has a benign trailing-edge stall character at $Re \sim 3.5 \cdot 10^5$; alternatives NACA 0015 (thicker, even more benign), SD8020. Cambered canard sections (e.g. NACA 2412) give more lift per chord but stall more sharply — reject for the gentle-stall priority. *Outputs:* Cl, α (2-D), Cl_{max} , Cm_0 , Cd_0 polar for main and canard. *Handoffs:* Phase 8 (Cl_{max} for V_{stall}), Phase 9 (canard polar), Phase 10 ($Cm-\alpha$). *Open assumptions:* Forced transition $x/c = 0.50$ is a conservative low-Re hedge against laminar-bubble drag (Selig 2003 §3.4). *Why now:* Cannot run before Re is estimated (Phase 3 product).

Phase 8 — Main-wing planform: AR, λ , sweep, S → V_{stall} . *Inputs:* Cl, α and Cl_{max} (Ph 7), V_{cruise} (Ph 3), MTOW. *Governing equations:*

- Lift-curve slope (3-D): Polhamus/Helmbold equation (USAF DATCOM §4.1.3.2): $C_{L\alpha} = 2\pi \cdot AR / [2 + \sqrt{(AR^2(1-M^2)(1 + \tan^2 \Lambda_c/2 / (1-M^2)) / (k^2 + 4))}]$, with $k = Cl, \alpha / 2\pi$. (Cited in Gudmundsson 2022 Eq. 9-72 as "the Polhamus equation ... presented on pp. 4.1.3.2-49 in [USAF DATCOM].")
- 3-D max lift: $Cl_{\text{max},3D} \approx 0.9 \cdot Cl_{\text{max}} \cdot \cos \Lambda_c / 4$ (Raymer 2018 Eq. 12.16; Roskam Part VI).
- Stall speed (DERIVED HERE): $V_{\text{stall}} = \sqrt{(2 \cdot (W/S) / (\rho_{\text{SL}} \cdot Cl_{\text{max},3D}))}$. Sea-level ρ because stall is most restrictive at low altitude.
- Oswald e for small UAVs: $e = 0.7-0.85$ (Gudmundsson Ch. 9). Ananda, Sukumar & Selig 2015, *Aerospace Science and Technology* Vol. 42, pp. 392-406 ("Measured aerodynamic characteristics of wings at low Reynolds numbers") report verbatim that "Oswald's

efficiency factors derived from the measurements ranged from a low value of 0.3 to more typical values near 0.8" for AR 2–5 wings at Re 60 000–160 000. **Caveat:** that study tested only AR 2–5, not the AR 6–8 range Bellona uses, so the 0.78 baseline is extrapolated; the 0.85 manned-aircraft upper bound is rejected as too optimistic for UAS structures.

UIUC Applied Aerodynamics (University of Illinois)

- Wing area from a chosen W/S that satisfies: (a) $V_{\text{stall}} \leq 1.0 \cdot V_{\text{min,transition}}$ (Stone 2008 transition-floor: $V_{\text{min}} \approx 0.8 \cdot V_{\text{stall_seal}}$ for safe hover-to-FW), (b) handling-quality $V_{\text{stall}} \leq 12$ m/s for the post-capture tow case. *Outputs:* S, b, $\bar{c}$, AR, λ , $C_{L\alpha}$, CL_{max} , 3D, V_{stall} , x_{ac_w} . *Handoffs:* Phase 9 (S for canard ratio), Phase 10 (geometry for scissor plot), Phase 11 (b for aileron span). *Open assumptions:* $\lambda = 0.4$ baseline retained; sweep zero for low-Re/SD7037. e refined after Re loop closes. *Why now:* Depends on both CL_{max} (Ph 7) and V_{cruise} (Ph 3). This is the *only* place where V_{stall} is derived — explicitly addressing the hard constraint that $V_{\text{stall}} \neq f(V_{\text{cruise}})$.

Phase 9 — Canard planform. *Inputs:* S (Ph 8), CL_{α_w} , x_{ac_w} (Ph 8), canard airfoil polar (Ph 7).

Governing equations:

- Volume coefficient: $V_c = (S_c \cdot l_c) / (S \cdot \bar{c})$ target 0.30–0.50 for canard configurations (Nicolai & Carichner 2010 Vol. I §11.6; Raymer 2018 Tab. 6.4 gives canard volume coefficients).
- Canard lift slope (DATCOM Polhamus): $C_{L\alpha_c} = 2\pi \cdot AR_c / [2 + \sqrt{(\dots)}]$, same formula as main wing.
- Canard-stall-first rule: $(Cl_{\text{max}_c} \cdot S_c) > (Cl_{\text{max}_w} \cdot S_w) \cdot (l_c / l_w)$ condition adapted from Roskam Part II Ch. 8 — equivalently in trim, the canard reaches its max-lift angle of attack before the wing. *Outputs:* S_c , AR_c , b_c , $\bar{c}_c$, l_c , V_c , $C_{L\alpha_c}$. *Handoffs:* Phase 10 (scissor plot). *Open assumptions:* l_c is constrained by fuselage length and rotor-boom geometry; treat as design variable in the scissor-plot loop. *Why now:* Depends on wing geometry from Phase 8.

Stability & Control

Phase 10 — Longitudinal static stability and scissor plot (canard form). *Inputs:* Wing & canard geometry, lift slopes, downwash/upwash gradients. *Governing equation* — CANARD neutral point, Phillips NASA TM-86694 (1985), Eq. 16: $x_{\text{np}}/\ell = [1 + (a_w(1 - \epsilon_{\alpha_c}) \cdot S_w) /$

$(a_c(1 + \epsilon_{\alpha_w}) \cdot S_c)]^{-1}$ where ℓ is the distance from canard AC to wing AC, a_c , a_w are the *isolated* lift-curve slopes, ϵ_{α_c} is the downwash gradient at the wing due to the canard, ϵ_{α_w} is the upwash gradient at the canard due to the wing. Because $0 \leq x_{\text{np}}/\ell \leq 1$, the neutral point always lies *between* the canard AC and the wing AC; increasing S_c or a_c moves x_{np} forward — the canard's destabilising contribution. Equivalent canonical form (Nelson 2nd ed. Eq. 2.42 adapted for canard sign): $x_{\text{np}}/\bar{c} = x_{\text{ac}_w}/\bar{c} - (C_{L\alpha_c}/C_{L\alpha_{AC}}) \cdot \eta_c \cdot V_c \cdot (1 + \epsilon_{\alpha_w})$ where the (–) sign in front of the canard contribution is the inverse of the conventional aft-tail (+) — this directly captures the *destabilising* forward shift. Caughey MAE 5070 Cornell course notes

(2011) Eq. 3.29 states this verbatim: " ΔNP is positive for conventional (aft tail) configurations, and negative for canard (forward tail) configurations." (nasa) (cornell)

- *Scissor plot*: plot allowable CG range vs $S_c \cdot l_c$ product:
 - *Forward CG limit*: set by maximum trimmable canard lift, $C_{L,c,max}$ — i.e. canard must produce enough nose-up moment to trim at landing/stall AoA. Equation: $x_{cg,fwd}/\bar{c} = x_{ac,w}/\bar{c} - (S_c \cdot C_{L,c,max} \cdot l_c) / (S \cdot \bar{c} \cdot C_{L,trim})$.
 - *Aft CG limit*: $x_{cg,aft}/\bar{c} = x_{np}/\bar{c} - SM_{min}$. $SM_{min} = 0.05$ (autopilot-assisted UAV — Beard & McLain 2012 §5.4, Gundlach 2014 §10.5); 0.10 recommended baseline.
- Outputs*: Allowable CG range; S_c sometimes revised back into Phase 9. *Handoffs*: Phase 11 (CG range $\rightarrow$ required $C_{m,\delta e}$ range), Phase 14 ($C_{m,\alpha}$ for dynamic check), Phase 15 (CG envelope drives equipment layout). *Open assumptions*: ε_α and ε_w from a vortex-lattice (e.g. AVL) prelim run or from DATCOM Fig. 4.4.1.

Phase 11 — Elevon sizing in FW mode (roll + pitch authority). *Inputs*: CG range and required $C_{m,\delta e}$ (Ph 10); roll-rate spec (mission-derived or MIL-F-8785C). *Governing equations*:

- Elevon effectiveness: $C_{m,\delta e} = -C_{L,\alpha} \eta_e \cdot (c_e/c) \cdot (S_e/S) \cdot (l_e/\bar{c}) \cdot \tau_e$, where τ_e is the flap-effectiveness parameter from Roskam Part VI Fig. 8.13.
- Roll authority: $C_{l,\delta a} = (2 \cdot C_{L,\alpha} \tau_e \cdot c_a/b) \cdot \int y_1^2 (c(y) \cdot y) dy / (Sb)$ (Nelson Ch. 3).
- Roll-rate steady-state: $p_{ss} = -C_{l,\delta a} \delta a \cdot V / (C_{l,p} \cdot b/2)$ (Roskam Part VII Ch. 10).
- Propwash dynamic-pressure factor: for canard tail-sitter, *forward* rotors wash over *aft* main wing elevons — $q_{local} = q_\infty \cdot (1 + 8 \cdot C_T / (\pi \cdot J^2))$ (from Stone 2008's momentum-theory development of $V_s = \sqrt{(V_\infty^2 + 2T/(\rho A))}$).
- Required roll rate: derive from mission. (a) Spiral-overshoot manoeuvre at bank $\leq 15^\circ$: need $p \geq \Delta\phi/\Delta t \approx 15^\circ/0.5 s \approx 30^\circ/s$ for soft correction; (b) gust-recovery for 5 m/s gust at $V_{cruise} \approx 18$ m/s: bank $\leq 30^\circ$, recovered in 1 s $\rightarrow p \geq 30^\circ/s$. (c) MIL-F-8785C §3.3.4 Class I Phase Cat. C Level 1 floor: roll-control commands "up to the magnitude which causes a 60-degree bank angle change in 1.7 T_d seconds" — equivalent steady roll rate ≈ 35 – $60^\circ/s$ depending on roll-mode time constant. **Take the MIL spec as floor: design for $p \geq 60^\circ/s$** to keep margin over mission-derived $30^\circ/s$. (Pku)
- Pitch authority: must trim full CG range across $V_{min,FW}$ to $V_{max,cruise}$. Critical case is forward-CG at low V (post-transition pull-up) — apply propwash boost q_{local} on elevons. *Outputs*: c_e/c , b_e/b , τ_e schedule, achievable p , achievable $\Delta\alpha_{trim}$. *Handoffs*: Phase 13 (FW endpoint of blending), Phase 14 ($C_{m,\delta e}$ for SP dynamics). *Open assumptions*: τ_e from DATCOM (Roskam Part VI Fig. 8.13). *Why now*: Needs CG range from Ph 10.

Phase 12 — Hover control authority (binding case for elevons). *Inputs*: Rotor positions and v_h (Ph 1, 2); I_{xx} , I_{yy} from preliminary Phase 15 mass distribution; required hover rates. *Governing equations*:

- *Pitch and roll via differential thrust*: $M_y = \Delta T \cdot d_x$ (longitudinal pair); $\Delta T/T_{\text{hover}}$ ratio for required $\omega = M/I_{yy}$. For Bellona X-config with $d_x \approx 0.6$ m and $I_{yy} \approx 8$ kg·m², a 5 % differential thrust gives $\omega_{\text{pitch}} \approx (0.05 \cdot 12.5g \cdot 0.6)/8 \approx 0.46$ rad/s². Required hover-pitch bandwidth ~ 2 rad/s (ADS-33E §3.3.5 hover/low-speed mode bandwidth) $\rightarrow$ need $\Delta T/T \approx 15$ –20 % capability.
- *Yaw via elevons in propwash*: $M_z = q_{\text{slip}} \cdot S_e \cdot C_{L,\delta e} \cdot \delta e \cdot l_z$, with $q_{\text{slip}}(\text{hover}) = T/A_{\text{disc}}$ (Stone 2008; momentum theory, $V_\infty = 0$; identical to $V_s^2 \rho/2$ with $V_s = \sqrt{2T/(\rho A)}$). Habel et al. (2025, "YawSitter," arXiv 2510.02968) Eqs. 4–7 give the same result with the development factor $B_d = 2$ placing the elevon in the fully-contracted far wake: $V_{s,i} = \sqrt{4T_i/(\rho \pi R_s^2)}$, $q_i = 2T_i/(\pi R_s^2)$. For Bellona DL ≈ 75 N/m² $\rightarrow V_{\text{slip}} \approx \sqrt{2 \cdot 75/0.660} \approx 15$ m/s $\rightarrow q_{\text{slip}} \approx 75$ Pa (at the disc) or up to ~ 150 Pa in the far wake ($B_d \rightarrow 2$). (arxiv)
- Required yaw rate in hover: target $\sim 30^\circ/\text{s}$ (ADS-33E Level 1 hover); equivalently, balloon-track heading correction at 100 m range, 1 m/s wind $\rightarrow$ need yaw bandwidth ≥ 0.5 rad/s.
- *Sizing*: compare δe span/chord required against Phase 11 result; **take the MAXIMUM**. For tail-sitter X-configs the hover-yaw case typically binds (Stone 1999 PhD thesis §5 "Configuration Design of a Canard Configured Tail-Sitter UAV Using Multidisciplinary Optimisation," Univ. Sydney; Bapst et al. IROS 2015 §III; Hochstenbach et al. on Wingtra). *Outputs*: Final elevon S_e, b_e, c_e ; required δe_{max} . *Handoffs*: Phase 11 (max-of comparison closes elevon spec), Phase 13. *Open assumptions*: I-tensor from Phase 15 prelim; ADS-33 used as rotorcraft-like spec during hover (Aeronautical Design Standard ADS-33E-PRF §3 — current US Army handling-quality spec for VTOL).

Phase 13 — Transition control blending. *Inputs*: Phase 11 elevon spec; Phase 12 differential-thrust limits; Phase 3 transition altitude. *Governing equations*:

- Mixer gain $\alpha(V)$: $\alpha = \text{clip}((V - V_{\text{blend_start}})/(V_{\text{blend_end}} - V_{\text{blend_start}}), 0, 1)$, with $V_{\text{blend_start}} = 0.5 \cdot V_{\text{stall}}$ and $V_{\text{blend_end}} = 1.2 \cdot V_{\text{stall}}$ (Stone & Clarke 2001 AIAC Paper 105; Stone & Wong 2004; Bapst et al. IROS 2015).
- Effective control = $(1-\alpha) \cdot u_{\text{diff_thrust}} + \alpha \cdot u_{\text{elevon}}$.
- Transition energy: $E_{\text{trans}} = \int T(t) \cdot V(t) dt$ over transition duration ~ 3 –7 s (Li et al. 2020, *IEEE Access*, "Transition Optimization for VTOL Tail-Sitter UAV" — collocation-method trajectory optimisation minimising energy with 3-DOF longitudinal model). *Outputs*: Transition duration, average T , transition energy (feeds Ph 5). *Handoffs*: Phase 5 (energy), Phase 14 (transition dynamics check). (IEEE Xplore)

Phase 14 — Dynamic stability (VERIFICATION). *Inputs*: All stability derivatives from Phases 10–11; I-tensor (Ph 15). *Governing equations*:

- Short-period: $\omega_{n,\text{sp}} = \sqrt{((C_{m,\alpha} \bar{q} \cdot S \cdot \bar{c})/I_{yy} + \dots)}$ (Etkin & Reid 1996 Ch. 6).
- Targets: $\omega_{n,\text{sp}} \geq 2$ rad/s, $\zeta_{\text{sp}} \in [0.35, 1.3]$ (MIL-F-8785C §3.2.2.1.1 Class I Level 1).

- Dutch roll, spiral: MIL-F-8785C §3.3.1.
- Phugoid: $\zeta_{ph} \geq 0.04$ (Level 1). *Decision:* Verification only — if a target is missed, flag for autopilot SAS design rather than re-sizing.

Mass & Closure

Phase 15 — Class II mass breakdown. *Inputs:* All geometric outputs. *Governing equations:*

- Wing mass: $m_w = (S \cdot W_{planform}/g) \cdot K_w$, with K_w from Roskam Part V Tab. 5.1 corrected $\times 0.7$ for UAV scale (Gundlach 2014 Ch. 19 cautions that manned-aircraft regression overestimates UAV structural mass by 30–50 %).
- Fuselage: Raymer Eq. 15.49 (light aircraft form) $\times 0.6$ UAV correction.
- Motors: $m_{motor} \approx 0.27 \text{ g/W} \cdot P_{cont}$ (Gundlach Tab. 9.3).
- Battery: Phase 5.
- Avionics, capture system (net 0.5 kg, tether 1.0 kg, reel 0.5 kg, hardpoint 0.3 kg). *Outputs:* m_{total} breakdown, CG location, I-tensor. *Handoffs:* Phase 16, Phase 10 (CG), Phase 14 (I-tensor).

Phase 16 — MTOW convergence (outer loop). *Procedure:* If $|MTOW_{new} - MTOW_{old}|/MTOW_{old} > 0.01$, set $MTOW_{next} = MTOW_{old} + \lambda \cdot (MTOW_{new} - MTOW_{old})$ with damping $\lambda = 0.4$ (Gundlach 2014 Ch. 19 recommends 0.3–0.5). *Stop:* $|\Delta MTOW|/MTOW < 1\%$ AND constraint diagram (Ph 6) still feasible AND dynamic stability (Ph 14) still satisfied.

DELIVERABLE 2 — N2 Dependency Matrix

Diagonal cells are the phase boxes; off-diagonal cells show the SPECIFIC quantity flowing from row→column. Italic entries are *inner-loop feedback* edges; **bold** entries are *outer-MTOW-loop* edges.

from↓ \ to→	P1	P2	P3	P4	P5	P6	P7	P8
P1 Prop	■	D, A_disc	—	n_max, D	—	—	—	—
P2 Hover P	—	■	P_hov(h), v_h	—	E_hov(t)	P/W_hov	—	—
P3 Mission	—	—	■	V_cruise	E_FW(t), γ, ROC	(V_cr, ROC)	V_cr, Re_est	V_cruise
P4 Prop/J loop	<i>D_revise</i>	—	<i>V_cruise feedback</i>	■	η_prop(seg)	—	—	—
P5 Energy/batt	—	—	—	—	■	(W/P)	—	—
P6 Constraint	—	—	—	—	—	■ verify	—	—
P7 Airfoil	—	—	—	—	—	—	■	Cl,α, Cl,max, Cd0
P8 Wing	—	—	—	—	—	V_stall, (W/S)	<i>Re_final</i> (Re loop)	■
P9 Canard	—	—	—	—	—	—	—	—
P10 Scissor	—	—	—	—	—	—	—	—
P11 Elevon FW	—	—	—	—	—	—	—	—
P12 Hover ctrl	—	—	—	—	—	—	—	—

from↓\to→	P1	P2	P3	P4	P5	P6	P7	P8
P13 Transition	—	—	—	—	E_trans	—	—	—
P14 Dyn stab	—	—	—	—	—	—	—	—
P15 Mass	—	—	—	—	—	—	—	—
P16 MTOW conv	MTOW	MTOW	MTOW	—	MTOW	—	—	MTOW

Four key iteration loops:

1. **MTOW outer loop** (bold): P16 → P1, P2, P3, P5, P8, P15. Damping 0.4.
2. **Re inner loop** (italic): P8 → P7. Final Re from converged S refines airfoil polars; iterate once (Selig 2003 notes $C_{l,max}$ varies <5 % across the band).
3. **CG-canard area loop** (italic): P10 → P9. If scissor-plot feasible region shrinks below 5 % MAC CG band, increase S_c .
4. **V_cruise-J loop** (italic): P4 → P3 (revise V_{cr}) or P4 → P1 (revise D). Typically resolved in one outer iteration if D is sized close to the disc-loading regression optimum.

DELIVERABLE 3 — Justified Assumption Table

Parameter	First guess	Valid range	Justification	Citation
FoM (figure of merit, hover)	0.70	0.65–0.75	Small electric UAV rotor at $DL \sim 75$ N/m ² ; below large-helicopter 0.75–0.80 because of blade-element inefficiencies at low Re	Leishman 2006 Ch. 2; Gundlach 2014 Tab. 7.4
η_{motor} (brushless out-runner)	0.90	0.85–0.93	Typical brushless DC at 100–2000 W in nominal RPM range	Gundlach 2014 Ch. 9 §9.4
η_{ESC}	0.95	0.93–0.97	Modern PWM ESC	Gundlach 2014 Ch. 9 §9.5

Parameter	First guess	Valid range	Justification	Citation
$\eta_{\text{prop cruise}}$	0.75	0.65–0.82	Fixed-pitch composite, $J = 0.4\text{--}0.8$	Gudmundsson 2022 Ch. 14 §14.6; UIUC propeller DB
$\eta_{\text{batt round-trip}}$	0.95	0.90–0.97	LiPo coulombic efficiency at moderate C-rate	Gundlach 2014 Ch. 8 §8.6
f_{usable} (battery DoD)	0.85	0.80–0.90	DoD limit for cycle-life > 200 cycles	Gundlach 2014 Ch. 8
T/W design	1.30	1.15–1.40	Stone & Clarke 2001 floor 1.15 for tail-sitter transition; 1.30 gives 13 % control margin	Stone & Clarke AIAC 2001 Paper 105; Stone PhD thesis 1999
C_{D0} (FW-VTOL class)	0.040	0.030–0.055	Small UAV with exposed rotors, struts, antennae	Gundlach 2014 Ch. 5 Tab. 5.2; Raymer 2018 Tab. 12.4
Oswald e	0.78	0.70–0.85	UAS-specific (NOT 0.85–0.95 manned). Ananda, Sukumar & Selig 2015 <i>Aerospace Sci. Tech.</i> Vol. 42 pp. 392–406 measured AR 2–5 wings at Re 60k–160k: "Oswald's efficiency factors ranged from a low value of 0.3 to more typical values near 0.8" (Illinois) — Bellona AR 7 sits at the upper end.	Gundlach 2014 Ch. 9; Ananda et al. 2015
Static margin SM	0.10 MAC	0.05–0.15	Autopilot-assisted UAV; lower bound permissible because SAS stabilises SP	Beard & McLain 2012 §5.4; Gundlach 2014 §10.5
Forced transition x/c	0.50	0.40–0.60	Conservative laminar-bubble hedge at Re $\sim 7 \cdot 10^5$	Selig 2003 VKI LRN-design notes §3.4
e_{batt} (LiPo pack)	200 Wh/kg	150–250 (upper bound of practice)	Pack-level (not cell), 4S–8S. EcoFlow Oct 2025: "150 to 250 Wh/kg for LiPo packs used in small multirotors." (EcoFlow) (ecoflow) timharper.net 2025:	EcoFlow 2025; timharper.net 2025

Parameter	First guess	Valid range	Justification	Citation
			"pack-level densities typically reach 150–200 Wh/kg." (Tim Harper)	
Disc loading	75 N/m ²	100–300 N/m ² for VTOL UAVs	Bellona MTOW/n/A_disc	Gundlach 2014 Ch. 7
Tip Mach limit	0.85	0.80 (wood) – 0.90 (thin composite)	Compressibility & noise — "Most aircraft designers aim to keep tip speed below Mach 0.85 (about 949 ft/s)" (liminfo)	Hibbs (Kitplanes "Prop Blade Mach"); Gudmundsson 2022 Ch. 14
Advance ratio J range	0.4–0.8	0.3–0.9	Fixed-pitch composite $\eta_{\text{prop}} > 0.7$	Gudmundsson 2022 Ch. 14; UIUC prop DB
q _{slip} /q _∞ in cruise (forward rotor over aft wing)	1.3–1.7	1.1–2.5	Stone 2008 momentum-theory form $1 + 8C_T/(\pi J^2)$ at $C_T \approx 0.05$, $J \approx 0.6$	Stone 2008 J. Aircr. 45(1) doi:10.2514/1.15705
Roll rate min	60°/s	45–90°/s	MIL-F-8785C §3.3.4 Class I Phase Cat C Level 1: "60-degree bank angle change in 1.7 T _d seconds" (Pku)	MIL-F-8785C §3.3.4
Pitch rate min (FW)	30°/s	20–60°/s	MIL-F-8785C §3.2.2 Cat C Level 1; mission-derived gust-recovery 5 m/s in 1 s	MIL-F-8785C §3.2.2
Yaw rate hover	30°/s	20–60°/s	ADS-33E Level 1 hover bandwidth	ADS-33E-PRF §3.3.5
Wing mass fraction (×UAV correction)	0.7·Roskam	0.6–0.85	UAV correction for Roskam Part V equations	Gundlach 2014 Ch. 19

Parameter	First guess	Valid range	Justification	Citation
Fuselage mass fraction (×UAV correction)	0.6·Raymer	0.5–0.75	Same — Raymer Eq. 15.49 corrected for UAV	Gundlach 2014 Ch. 19; Raymer Ch. 15
Motor mass/power	0.27 g/W cont.	0.20–0.35	Out-runners 100–1000 W	Gundlach 2014 Tab. 9.3
Battery mass fraction	30 % MTOW	25–40 %	Bellona high-energy mission	Derived from Phase 5 + e_batt
Avionics mass	1.5 kg	1.0–2.5 kg	Pixhawk-class flight controller + GNSS/IMU/datalink + power module	Gundlach Ch. 11–12

DELIVERABLE 4 — Python Code Outline

python

```
"""
```

```
Bellona DSE Group 22 – Detailed-Design Sizing Skeleton.
```

```
Dependencies: numpy, matplotlib only.
```

```
"""
```

```
import numpy as np
import matplotlib.pyplot as plt
from dataclasses import dataclass
from typing import Tuple, Dict, Callable
```

```
# ----- HELPERS -----
```

```
def isa(h: float) -> Tuple[float, float, float, float]:
```

```
    """ISA 1976 atmosphere (troposphere only, 0-11 km).
```

```
    In:  h [m]
```

```
    Out: rho [kg/m^3], mu [Pa·s], a [m/s], T [K]
```

```
    Ref: U.S. Standard Atmosphere 1976."""
```

```
    ...
```

```
def disc_loading_regression(MTOW_N: float, n_rotors: int, DL_target: float
                           ) -> float:
```

```
    """First-cut prop diameter from disc-loading regression.
```

```
    In:  MTOW_N [N] (Ph16), n_rotors, DL_target [N/m^2] (Ph1, Tab 7.4 Gundlach)
```

```
    Out: D_prop [m]
```

```
    Eq:   $D = \sqrt{4 \cdot T_{\text{rotor}} / (\pi \cdot DL)}$  – Gundlach 2014 Eq. 7.18 form
```

```
    Loop: MTOW outer."""
```

```
    ...
```

```
# ----- PHASE 1 -----
```

```
def phase1_propeller(MTOW_N: float, T_over_W: float, n_rotors: int,
                     DL_target: float, a_sound: float, M_tip_max: float = 0.85
                     ) -> Dict:
```

```
    """Physical propeller sizing.
```

```
    In:  MTOW [N], T/W, n_rotors, DL_target [N/m^2], a [m/s] (Ph0 ISA), M_tip_max
```

```
    Out: {'D_prop' [m], 'A_disc' [m^2], 'n_max' [rev/s], 'V_tip' [m/s]}
```

```
    Eq:   $D = \sqrt{4T / (\pi \cdot DL)}$ ;  $n_{\text{max}} = M_{\text{tip\_max}} \cdot a / (\pi \cdot D)$ 
```

```
    Ref: Gudmundsson 2022 Ch.14; Hibbs Kitplanes "Prop Blade Mach"
```

```
    Loop: MTOW outer; J-feedback inner (Ph4)."""
```

```
    ...
```

```
# ----- PHASE 2 -----
```

```
def phase2_hover_climb_power(T_per_rotor: float, A_disc: float, rho: float,
```

```

        V_climb: float, FoM: float, eta_motor: float,
        eta_ESC: float) -> Dict:
    """Actuator-disc induced velocity, hover & axial-climb power.
    In:  T [N], A_disc [m^2], rho [kg/m^3] (Ph0), V_c [m/s], FoM, eta_motor, eta_ESC
    Out: {'v_i', 'P_shaft', 'P_elec'} [m/s, W, W]
    Eq:  v_i = -V_c/2 + sqrt((V_c/2)^2 + T/(2·rho·A))  - Leishman 2006 Eq. 2.91
        P_shaft = T(V_c + v_i)/FoM                    - Leishman Eq. 2.96
    Loop: MTOW outer."""
    ...

# ----- PHASE 3 -----

def phase3_mission_optimise(MTOW_N: float, S_guess: float, CD0: float, AR: float,
                            e: float, h_target: float, range_target: float,
                            t_budget: float, t_hover: float,
                            P_hover_fn: Callable, isa_fn: Callable = isa
                            ) -> Dict:
    """Derive V_cruise, gamma, ROC, h_transition by minimising total energy.
    In:  MTOW [N], S [m^2] (Ph8 estimate), CD0, AR, e (Ph8 estimate),
        h_target=6000 m, range_target=6000 m, t_budget=600 s, t_hover=300 s,
        P_hover_fn(h)->W (Ph2)
    Out: {'V_cruise', 'gamma', 'ROC', 'h_tr', 'Re_estimate'}
    Eq:  V_Pmin = sqrt(2(W/S)/rho · sqrt(K/(3·CD0)))  - Raymer 2018 Ch.17
        ROC = (P_avail - P_req)/W                    - Raymer Eq. 17.22
    Method: grid-search h_tr in [50, 2000] m, inner 1-D minimise V over P_req(V).
    Loop: MTOW outer; V-J feedback from Ph4."""
    ...

# ----- PHASE 4 -----

def phase4_J_coupling(V_cruise: float, n_rev_s: float, D_prop: float,
                      J_min: float = 0.4, J_max: float = 0.8) -> Dict:
    """Verify advance ratio in efficient band; signal D-revise if not.
    In:  V_cruise [m/s] (Ph3), n [rev/s] (Ph1), D [m] (Ph1)
    Out: {'J', 'in_band' (bool), 'D_recommend'}
    Ref: Gudmundsson 2022 Ch.14 §14.6; UIUC propeller DB.
    Loop: V_cruise-J inner loop with Ph3 (or Ph1 D-revise)."""
    ...

# ----- PHASE 5 -----

def phase5_energy_battery(segments: Dict[str, Tuple[float, float]],
                           eta_batt: float = 0.95, f_usable: float = 0.85,
                           e_batt_Wh_kg: float = 200.0) -> Dict:

```



```
"""Mission energy timeline, battery mass.
```

```
In:  segments={'name': (P[W], dt[s])} – VTOL climb, transition, FW climb,
      hover, tow, return cruise, descent, reserve. (Ph2, Ph3, Ph13)
```

```
Out: {'E_total_Wh', 'm_batt_kg', 'P_motor_cont', 'P_motor_peak'}
```

```
Eq:  E = sum(P*dt)/(eta_batt*f_usable) – Gundlach 2014 Eq. 8.4
```

```
Loop: MTOW outer."""
```

```
...
```

```
# ----- PHASE 6 -----
```

```
def phase6_constraint_diagram(W_S_range: np.ndarray, W_P_range: np.ndarray,
                              V_cruise, V_stall, ROC, gamma, rho_SL, rho_6km,
                              CL_max, CD0, AR, e, eta_prop, T_W_floor) -> None:
    """Plot W/S vs W/P with stall, cruise, climb, hover, transition-floor lines.
    Verification step (post Ph3, Ph5, Ph8). Not a sizing driver.
    Ref: Raymer 2018 Ch.5 §5.3."""
```

```
...
```

```
# ----- PHASE 7 -----
```

```
def phase7_airfoil_xfoil(Re_main: float, Re_canard: float,
                          x_transition: float = 0.50) -> Dict:
    """XFoil wrapper: evaluate SD7037 + NACA 0012 + candidates.
    In:  Re_main, Re_canard (Ph3+Ph8 estimate), forced transition x/c (Selig 2003).
    Out: {'main': {'cl_a', 'cl_max', 'cd0', 'cm0', 'stall_char'},
          'canard': {...}}
    Ref: Drela XFOIL User Guide; Selig 2003 VKI LRN lecture notes.
          SD7037 Cl/Cd_max ≈ 75 @ Re=2e5, ≈ 99 @ Re=5e5 (UIUC database).
    Loop: Re inner loop with Ph8."""
```

```
...
```

```
# ----- PHASE 8 -----
```

```
def phase8_wing_planform(MTOW_N: float, cl_max_2D: float, cl_a_2D: float,
                          rho_SL: float, V_stall_target_max: float,
                          AR_guess: float = 7.0, taper: float = 0.4,
                          sweep_c4: float = 0.0, e: float = 0.78) -> Dict:
    """Wing planform; DERIVE V_stall from CL_max,3D and chosen W/S.
    In:  MTOW [N], cl_max,2D, cl_a,2D (Ph7), rho_SL,
          V_stall_target_max [m/s] (handling-quality + Stone transition floor),
          AR, taper, sweep (design choices).
    Out: {'S', 'b', 'c_bar', 'AR', 'CL_a', 'CL_max_3D', 'V_stall', 'x_ac_w'}
    Eq:  CL_a = 2*pi*AR/(2 + sqrt(AR^2(1-M^2)(1+tan^2 Lambda_c2/(1-M^2))/k^2+4))
          – Polhamus / DATCOM §4.1.3.2
```

```

        CL_max_3D = 0.9·c_l_max·cos(Lambda_c4)      - Raymer 2018 Eq. 12.16
        V_stall   = sqrt(2(W/S)/(rho_SL·CL_max_3D))

Loop: Re inner (Ph7); MTOW outer.""
...

# ----- PHASE 9 -----

def phase9_canard(S_w: float, c_bar_w: float, x_ac_w: float, l_c: float,
                  V_bar_c: float = 0.40, AR_c: float = 5.0,
                  taper_c: float = 0.5) -> Dict:
    """Canard planform from volume coefficient.
    In:  S_w, c_bar_w, x_ac_w (Ph8), l_c, V_c target (Nicolai §11.6).
    Out: {'S_c', 'b_c', 'c_bar_c', 'AR_c', 'l_c', 'CL_a_c'}
    Eq:  S_c = V_c · S_w · c_bar_w / l_c          - Nicolai 2010 Eq. 11.7
        CL_a_c via Polhamus (same as Ph8).
    Loop: CG-canard inner loop with Ph10."""
    ...

# ----- PHASE 10 -----

def phase10_scissor_canard(S_w, c_bar_w, x_ac_w, CL_a_w,
                           S_c, l_c, CL_a_c, eps_alpha_c, eps_alpha_w,
                           CL_c_max, CL_trim, SM_min: float = 0.10) -> Dict:
    """Longitudinal scissor plot using CANARD neutral-point formulation.
    In:  wing & canard geometry/derivs (Ph8, Ph9), downwash/upwash gradients,
        max canard lift coefficient, design SM_min (Beard&McLain 2012; Gundlach 201
    Out: {'x_np_over_c', 'x_cg_fwd_over_c', 'x_cg_aft_over_c', 'CG_range_pct'}
    Eq:  x_np/ell = [1 + a_w(1-eps_c)·S_w / (a_c(1+eps_w)·S_c)]^-1
                                                - Phillips NASA TM-86694 Eq.16

    Equivalent canonical (Nelson-style canard):
    x_np/c̄ = x_ac_w/c̄ - (C_La_c/C_La_AC)·eta_c·V_c·(1+eps_w)
    Caughey MAE5070 Eq.3.29: canard contribution is NEGATIVE (forward).
    Loop: CG-canard inner (revises S_c, Ph9)."""
    ...

# ----- PHASE 11 -----

def phase11_elevon_FW(S_w, b_w, c_bar_w, CL_a_w, V_cruise, V_stall,
                      CG_range, q_slipstream_ratio: float = 1.5,
                      p_required_rad_s: float = np.deg2rad(60),
                      Cm_de_required: float = 0.5) -> Dict:
    """FW-mode elevon: pitch trim + roll rate.
    In:  wing geom (Ph8), CG range (Ph10), q_slip/q_inf factor in cruise (Stone 2008
    Out: {'c_e_over_c', 'b_e_over_b', 'tau_e', 'Cm_de', 'Cl_da', 'p_achievable'}

```

Eq: $C_{m_de} = -C_{L_a} \cdot \eta_{e_e} \cdot (c_{e_e}/c) \cdot (S_e/S) \cdot (l_{e_e}/\bar{c}) \cdot \tau_{e_e}$
 $p_{ss} = -C_{l_da} \cdot \delta_{a_a} \cdot V / (C_{l_p} \cdot b/2)$ — Nelson Ch.3
 $q_{local} \text{ on aft-wing elevons} = q_{inf} \cdot (1 + 8 \cdot C_T / (\pi \cdot J^2))$ — Stone 2008
 Floor: MIL-F-8785C §3.3.4 Class I Level 1 $\rightarrow p \geq 60^\circ/s$

Loop: max-of with Ph12."""

...

----- PHASE 12 -----

```
def phase12_hover_control(T_hover_per_rotor: float, A_disc: float, rho: float,
                          d_x_rotor: float, d_y_rotor: float,
                          I_xx: float, I_yy: float, I_zz: float,
                          S_e_FW: float, b_e_FW: float, l_z_elev: float,
                          omega_dot_required: float = 2.0,
                          yaw_rate_required: float = np.deg2rad(30)) -> Dict:
    """Hover pitch/roll via differential thrust; yaw via elevon-in-propwash.
    In:  rotor thrust & geometry (Ph1), I-tensor (Ph15 prelim),
         FW elevon size (Ph11), required hover-rate specs (ADS-33E Level 1).
    Out: {'dT_over_T', 'b_e_yaw_required', 'S_e_yaw_required', 'binding_case'}
    Eq:  M_y = dT · d_x;  omega_dot = M/I
         q_slip(hover) = T/A_disc — Stone 2008 (V_inf=0)
         V_slip(hover) = sqrt(2T/(rho·A_disc)) — Leishman Eq.2.42 → Stone
         M_z_yaw = q_slip · S_e · CL_de · delta_e · l_z
    Decision: take MAX of (Ph11.b_e, Ph12.b_e_yaw_required)."""
    ...
```

----- PHASE 13 -----

```
def phase13_transition_blending(V_stall: float,
                                V_blend_start_frac: float = 0.5,
                                V_blend_end_frac: float = 1.2) -> Dict:
    """Control mixer schedule from hover (diff thrust) to FW (elevon).
    In:  V_stall (Ph8); blending fractions (Stone & Clarke 2001).
    Out: {'alpha_blend_fn', 'V_blend_start', 'V_blend_end',
         'E_transition_estimate', 't_transition'}
    Ref: Stone & Clarke AIAC 2001 Paper 105; Stone & Wong 2004; Bapst IROS 2015;
         Li et al. IEEE Access 2020 (collocation-method optimisation)."""
    ...
```

----- PHASE 14 -----

```
def phase14_dynamic_stability(Cm_a, Cm_q, Cm_a_dot, CL_a, I_yy, I_xx, I_zz,
                               S_w, c_bar_w, b_w, V_cruise, rho_cruise) -> Dict:
    """Short-period, phugoid, Dutch-roll, spiral — VERIFICATION ONLY.
```

```

In:  all stability derivatives & inertias (Ph10, 11, 15); flight cond (Ph3).
Out: {'omega_sp', 'zeta_sp', 'omega_ph', 'zeta_ph', 'omega_dr', 'zeta_dr',
      'T_spiral', 'level_meets_8785C'}
Ref: Etkin & Reid 1996 Ch.6; MIL-F-8785C §3.2, §3.3.
...

```

```
# ----- PHASE 15 -----
```

```

def phase15_mass(S_w, S_c, fuselage_length, P_motor_cont_W, m_battery_kg,
                 m_payload_kg: float = 5.0, m_capture_kg: float = 2.3,
                 UAV_correction_wing: float = 0.70,
                 UAV_correction_fuse: float = 0.60) -> Dict:
    """Class II mass breakdown with UAV correction factors.
    In:  geometric outputs, P_cont (Ph5), m_batt (Ph5).
    Out: {'m_wing', 'm_canard', 'm_fuse', 'm_prop', 'm_motor', 'm_ESC',
          'm_batt', 'm_avionics', 'm_payload', 'm_capture',
          'MTOW_kg', 'x_CG', 'I_tensor'}
    Eq:  m_wing  = K_w · f(S,W) [Roskam Part V Tab. 5.1] · UAV_correction
          m_fuse  = Raymer Eq.15.49 · UAV_correction
          m_motor = 0.27 g/W · P_cont                      – Gundlach 2014 Tab. 9.3
    Ref: Gundlach 2014 Ch.19 cautions 30-50 % overestimate without UAV correction.
    ...

```

```
# ----- PHASE 16 -----
```

```

def phase16_mtow_converge(MTOW_old: float, MTOW_new: float,
                          damping: float = 0.4, tol: float = 0.01
                          ) -> Tuple[float, bool]:
    """Outer-loop MTOW convergence with damping (Gundlach 2014 Ch.19).
    Returns (MTOW_next, converged_flag)."""
    delta = MTOW_new - MTOW_old
    MTOW_next = MTOW_old + damping * delta
    converged = abs(delta) / MTOW_old < tol
    return MTOW_next, converged

```

```
# ----- ORCHESTRATION -----
```

```

def converge_design(MTOW_seed_kg: float = 50.0, max_iter: int = 30,
                   tol: float = 0.01, damping: float = 0.4) -> Dict:
    """Outer MTOW loop orchestrating all 16 phases.
    Sequence (dependency-graph order, NOT constraint-diagram-first):
        0. ISA tables
        1. Phase 1  prop diameter
        2. Phase 2  hover/climb power

```

```

3. Phase 3 mission optimise (V_cruise, gamma, h_tr DERIVED here)
4. Phase 4 J coupling – inner V_cruise-J loop with Ph3
5. Phase 7 airfoil – inner Re loop seeded from Ph3 V_cruise
6. Phase 8 wing – V_stall DERIVED here from CL,max,3D and W/S
7. Phase 7' Re-loop refine (one iteration)
8. Phase 9 canard
9. Phase 10 scissor (canard form) – inner CG-canard loop with Ph9
10. Phase 11 FW elevon
11. Phase 12 hover elevon – MAX-of with Ph11 closes elevon spec
12. Phase 13 transition blending
13. Phase 5 energy / battery
14. Phase 15 mass – produces new MTOW
15. Phase 16 converge check (damping 0.4)
16. Phase 6 constraint diagram VERIFY (final)
17. Phase 14 dynamic stability VERIFY (final)
"""
MTOW = MTOW_seed_kg
history = []
for it in range(max_iter):
    # ... call sequence above ...
    MTOW_new = ... # from phase15
    MTOW, conv = phase16_mtow_converge(MTOW, MTOW_new, damping, tol)
    history.append(MTOW)
    if conv:
        break
# final verification
phase6_constraint_diagram(...)
phase14_dynamic_stability(...)
return {'MTOW': MTOW, 'history': history, ...}

```

Recommendations (staged)

1. **Immediately (week 1 of detailed design):** Code Phases 1, 2, 3 as standalone modules and verify against the midterm numbers as a sanity check — Phase 3 should reproduce a V_{cruise} close to (but not identical to) the midterm assumption; Phase 8 then derives V_{stall} *independently* and must give a value that satisfies handling QoS ($V_{stall} \leq 12$ m/s recommended).
2. **Week 2:** Run the Re inner loop (Phase 7 ↔ Phase 8) twice. Trigger to re-evaluate airfoil choice: if Re_{final} differs from the seed by more than 30 %, re-run XFOIL. SD7037

Cl/Cd_{max} is strongly Re -dependent (~ 75 at $Re=2 \cdot 10^5$ vs ~ 99 at $Re=5 \cdot 10^5$, UIUC database), so this iteration directly affects the cruise drag estimate.

3. **Week 3:** Phase 10 with the *canard* neutral-point equation. Benchmark threshold: if the resulting allowable CG range is below 10 % MAC, increase S_c by 10 % and rerun Phase 9 → Phase 10 (CG-canard inner loop).
 4. **Week 3-4:** Phases 11 and 12 in parallel. Decision threshold: if Phase 12 requires more elevon span than Phase 11, *take Phase 12's value* and verify the FW roll-rate margin grows above the MIL-F-8785C $60^\circ/s$ floor.
 5. **Week 4:** Phase 5 energy timeline and Phase 15 mass; then Phase 16 outer loop.
Convergence benchmark: $|\Delta MTOW|/MTOW < 1\%$ in ≤ 8 iterations with damping 0.4. If oscillation is observed (alternating overshoot), reduce damping to 0.3.
 6. **Week 5:** Run Phase 6 (constraint-diagram verification) and Phase 14 (dynamic-stability verification) as final checks. If either is missed, return to the phase that produces the deficient quantity (NOT to Phase 6).
 7. **Threshold for autopilot vs airframe trade:** If Phase 14 short-period $\zeta < 0.35$ but $\omega_{n,sp} > 2$ rad/s, accept and design a SAS (Beard & McLain Ch. 6); if $\omega_{n,sp} < 2$ rad/s, the airframe is too soft — return to Phase 10 and tighten SM to 0.10 minimum.
-

Caveats

- **q_{slip}/q_∞ at hover is mathematically singular** ($V_\infty \rightarrow 0$). For hover-yaw sizing the physically meaningful quantity is the absolute slipstream dynamic pressure $q_{slip}(\text{hover}) = T/A_{disc}$ (disc loading), not a ratio. Stone 2008 reports velocity ratios $V_s/V_\infty \approx 1.4\text{--}2.5$ in transition but no single "hover ratio."
- **The Phillips canard NP formula (TM-86694 Eq. 16)** assumes inviscid linear aerodynamics with first-order downwash/upwash; for the SD7037/NACA 0012 pair at Bellona's Re it should be cross-checked with a vortex-lattice run (AVL) before freezing the scissor plot.
- **The Ananda, Sukumar & Selig 2015 Oswald-efficiency data** were measured for AR 2-5 wings at Re 60 000-160 000 — *not* AR 6-8 at $Re \sim 8 \cdot 10^5$. The $e = 0.78$ baseline for Bellona is therefore extrapolated upward from the "more typical values near 0.8" reported in that study and should be re-checked with a vortex-lattice run at the converged Bellona geometry. Manned-aircraft $e = 0.85\text{--}0.90$ is rejected because it ignores the higher fuselage-interference fraction at UAS scale. (Illinois) (UIUC Applied Aerodynami...)
- **MIL-F-8785C Class I** is the closest US fixed-wing manned spec to a Bellona-class UAV; **ADS-33E-PRF** is used for hover/low-speed because it is the only Western standard that quantifies rotorcraft-like hover bandwidth. Neither is a UAV standard — for production, refer to NATO STANAG 4671 (UAV airworthiness) and the TU Delft Group 22 internal handling-quality table.

- **e_{batt} = 200 Wh/kg pack-level is at the *upper bound* of current LiPo practice** (EcoFlow Oct 2025: 150–250 Wh/kg for small-multirotor LiPo packs; timharper.net 2025: 150–200 Wh/kg typical after cooling/structure/safety overhead). If the hover phase demands > 5C continuous, derate to 170 Wh/kg. (EcoFlow + 2)
- **The midterm baseline's circular V_{stall} = f(V_{cruise}) must be removed at the first iteration;** the workflow above forces V_{stall} to be derived in Phase 8 from CL,max,3D and W/S, where W/S is itself constrained by V_{stall} ≤ V_{stall_max} and by the Phase 3 trajectory-energy optimum.
- **Manned-aircraft weight equations overestimate UAV mass by 30–50 %** (Gundlach 2014 Ch. 19); the UAV correction factors (0.7 wing, 0.6 fuselage) used in Phase 15 are first-cut and must be back-checked once a structural FEA is available in a downstream phase.
- **The Giguère & Selig 1997 roughness-insensitivity claim** ("the Cl,max of the ... SD7037 airfoils are not particularly sensitive to roughness," *Wind Engineering* Vol. 21 No. 6) applies specifically to Cl,max — *not* to drag or L/D. The cruise-drag estimate from the SD7037 polar can still be sensitive to bug/dust contamination on the leading edge, particularly at Re < 3·10⁵. (UIUC Applied Aerodynami...)